

# AI/ML Research for ProofRoad

Primary project source: ProofRoad v2, SIH 2026 PS 26007, uploaded by the user. The document describes ProofRoad as an engineering proposal and explicitly says it is not a safety certification. Page references below refer to the uploaded ProofRoad v2 document.

## 1. Where AI belongs

ProofRoad intentionally keeps the safety decision formula-based. AI can assist with object naming, fog trends and analytics, while the decision to slow down comes from the stopping-distance calculation (p. 14).

This creates a useful separation between probabilistic perception/analytics and deterministic safety-state logic.

## 2. Candidate AI tasks

Object classification from RGB/thermal data.

Multi-sensor association assistance.

Fog/dust classification as an advisory signal.

Near-miss and warning-pattern analytics.

Fleet-level simulation and scenario analysis.

Future fog trend forecasting, but only after collecting sufficient site data.

## 3. What AI should not control in the prototype

AI should not declare the road clear.

AI should not override the radar safety limit.

AI should not clear a hazard based on a confidence percentage.

AI should not convert a missing sensor into a higher safe speed.

AI should not directly brake a real truck in the hackathon prototype.

## 4. Data strategy

The project needs a labelled dataset containing clear/fog/dust conditions, obstacle classes, range, sensor health and route context. Because mine-specific fog data is not established in the ProofRoad document, synthetic augmentation should be labelled as synthetic and should not be presented as field validation.

A practical prototype can use controlled obstacles and thermal targets to validate the fusion logic while collecting structured logs for later model training.

## 5. Evaluation

Classification: precision, recall, confusion matrix and per-class performance.

Detection: range error, latency, false-alarm rate and missed-detection rate.

Safety logic: number of false-clear events must be zero in the designed test set; every uncertainty/failure case should map to a conservative state.

Human interface: alert count per minute, escalation behavior and response time should be studied later in shadow mode, as ProofRoad itself states that hackathon testing cannot establish human-factors results (p. 8).